

Lesson 4: Matrix algebra II

Learning outcomes: In this lesson, you will learn to

- Obtain the determinant, cofactors and adjoint of a square matrix
- Obtain the inverse of a non-singular matrix
- Solve a set of linear equations using inverse matrices

Lesson Activities:

Read the lecture notes:

- slides 35-51 from [Matrix algebra I slides](#)
- slides 1-10 from [Matrix algebra II slides](#)

Watch the lesson recordings:

- [1 recap of previous class](#)
- [2 Inverse of a square matrix](#)
- [3 Example inverse of a 3x3 matrix](#)
- [4 More examples and properties of inverses](#)
- [5 Introduction to systems of linear equations](#)
- [6 Solving systems of linear equations inverse matrix method](#)
- [session notes recap of previous session.pdf](#)

Do the follow-up exercises in the lecture notes

Note: The detailed video on how to find 3x3 determinants can be found [here](#).

Check Yourself:

Complete the Problem set

- [Problem Set 3](#)
- [Problem Set 3 answers/solutions](#)

Extra:

Check out the video below on how to derive the method of getting the inverse of a 2x2 matrix.

[Deriving the formula of the inverse of a 2x2 matrix](#)

Readings:

Read the following content from the [Core Textbook](#)

Prog 5 Pgs 499-502

Read the following content from [Core Textbook 2](#)

Chapter 13 pgs 576-603